IBM Project 1 - Validation and AI-use Record

Prepared: October 14, 2026, America/Indiana/Indianapolis
Model repository baseline: `0e3981bd1709ad0c35ef3eff8583c1d3afd0ca02`
Valuation cutoff: September 1, 2026, after close
Decision: Watch / Defer IBM common stock. This is a validation record, not a replacement investment decision.

1. Final model and execution environment

The final connected model uses USD millions except per-share data. It was independently executed for this record with the repository's `python3` command on macOS (Darwin 24.6.0 arm64), Python 3.14.7. The project requires Python 3.10+ and the standard library only.

Current calculated output	Result
IBM reference close	$231.40
Low / base / high DCF	$145.1346 / $182.2876 / $224.2811 per share
Base WACC / operating EV	7.0708% / $220,096.1m
Terminal PV share	78.0637%
September 1 cash / common equity	$9,280.5m / $173,665.4m
FY2026 company-style FCF proxy / remaining gap	$10,918.2m / $(4,781.8)m
Reverse-DCF terminal growth	3.8150%, holding 15% terminal incremental ROIC constant

2. Sources and SBC treatment

Item	Source and treatment
H1 2026	IBM Q2 2026 Form 10-Q, filed July 23, 2026: June 30 balance sheet, H1 results, $7,766m CFO, and $1,004m SBC.
FY2025	IBM FY2025 Form 10-K, filed February 24, 2026: FY2025 history and $1,715m SBC; H1 2025 $842m implies H2 2025 $873m.
SBC schedule	H2 2026 $938.5m: midpoint of reported H1 2026 $1,004m and historical H2 2025 $873m. FY2027-30 $1,877m each year, provisional and held flat.
Management FCF definition	$7,766m GAAP CFO - $2,264m Financing-receivable adjustment - $743m net capital investment = $4,759m, rounding to reported $4.760bn. Consolidated CFO, management FCF, and operating FCFF are distinct.

Independent historical-anchor check: IBM's Q2 2026 Form 10-Q segment revenue reconstructs to $33,079m (Software $14,813m + Consulting $10,599m + Infrastructure $7,161m + Financing $406m + Other $100m), equal to the model H1 anchor; difference $0m. See `docs/independent-source-check.md`.

Forecast SBC is assumed embedded in forecast operating-cost ratios. The model adds it once to consolidated CFO as a noncash adjustment and credits it once to APIC/equity. It is not a cash share issuance and is not added again to operating FCFF. The repair raises September 1 cash by $321.3m, or about $0.34 per share, but leaves operating FCFF and operating EV unchanged. Future dilution remains unresolved.

3. Validation register: independently executed October 14, 2026

Check	Expected result	Actual result and evidence
Automated suite	35 passing tests	**PASS** - 35 passed, 0 failed; 30-test baseline plus five independent high-risk checks; `python3 -m unittest discover -s tests -v`
Published Week 3 known-answer case	All 12 outputs reconcile; 11% WACC decreases value and 10% restores baseline	**PASS** - all 12 within tolerance; $23.4140/share at 11%; $27.4974/share restored at 10%. See `docs/week3-known-answer-reconciliation.md`.
Three-statement articulation	Balance-sheet and cash-flow gaps equal zero	**PASS** - FY2026-30 articulate within floating-point rounding.
Historical reconciliation	Opening, revenue, pretax, and H1 FCF reconcile	**PASS** - 0 / 0 / 0.0 bridge gaps; $4,759m H1 FCF rounds to $4,760m.
Enterprise-to-equity bridge	$182.2876 base value	**PASS** - $220,096.1m EV, $2,740.5m excess cash/securities, $173,665.4m common equity, 952.7m shares.
WACC and terminal direction	Value falls as WACC rises and rises with growth/ROIC	**PASS** - monotonicity and terminal-reinvestment tests passed.
Peer arithmetic	Separate peer bridge consistency	**PASS** - peer implied-value reconciliation test passed.
Scenarios	Low < base < high	**PASS** - $145.13 < $182.29 < $224.28.
September 1 stub	121/184 of H2 FCFF, from September 2	**PASS** - $4,791.1m residual FCFF.
Deliberate failure and recovery	$1m FY2027 cash break fails safely, normal model recovers	**PASS** - expected `$-1.0000m` balance-sheet failure; normal suite remained articulated.
SBC controls	One CFO add-back/APIC credit; no H1/H2 double count; no FCFF add-back	**PASS** - four dedicated SBC tests passed.

High-risk extension: a separate, read-only calculation path independently tied out the 121/184 stub, WACC, ROIC-based terminal reinvestment, operating EV ($220,096.1m), equity bridge ($173,665.4m), and $182.287630/share. It also recomputed the H1 company-FCF definition ($7,766m - $2,264m - $743m = $4,759m, rounded to $4,760m), tested H1/H2 period separation, and rejected invalid terminal-reinvestment and operating-cash bridge inputs. See `docs/high-risk-validation-evidence.md` and `docs/fy2026-fcf-source-reconciliation.md`.

`python3 model/ibm_phase3_model.py --write-output --demo-failure` printed the expected safe failure and regenerated output. A local workbench API request on port 8770 returned `statement_articulation: PASS` and base value `$182.287630`.

> **Historical versus task-run evidence.** This task did not repeat a fresh GitHub download. The October 14 fresh-download/launcher evidence in `docs/cold-run-log.md` is student-observed pre-SBC evidence: the launcher selected port 8768, the base was $181.95, and 24 tests were recorded. It is not represented as a final-model task-run result.

Complete current test list: the prior 30 tests above plus `test_independent_dcf_ties_out_without_production_dcf_call`, `test_period_correct_management_fcf_reconciliation`, `test_invalid_terminal_reinvestment_is_rejected`, `test_invalid_operating_cash_bridge_reserve_is_rejected`, and `test_material_consulting_shock_articulates_and_moves_value_down`; all 35 returned `ok`.

4. Historical Locked Changed-Input Record

The original locked text files and four screenshots remain unchanged in `docs/`. They document a pre-SBC experiment, not the current $182.29 baseline.

Historical field	Preserved record
Prediction / execution	October 10, 2026, 12:39pm, AI off / 12:53pm.
Input	Software revenue growth: 0pp to +2pp for FY2027-30.
Prediction	Revenue, operating profit, FCFF, and intrinsic value would rise; Watch / Defer would not change.
Observed result	$181.95 to $190.31/share; FY2030 EBIT $15,279.2m; FCFF $14,152.8m; WACC 7.07%.
Decision effect	Prediction direction matched. Value increased $8.36/share (4.6%) but remained below $231.40; Watch / Defer remained.
Provenance	Original predecessor repository prediction commit `7359ba2` is reported but unavailable here. Archived BEFORE commit `4591725`, timestamped 18:21:56 -04:00, is a later preservation event, not the original 12:39pm prediction time. Results/screenshots commit: `2f2384a`.

4A. Supplemental prospective Locked Changed-Input Record

This current-model test is supplemental, not a replacement for the October 10 record. The student declared that Part A was authored independently with generative AI assistance and autocomplete off. Its committed BEFORE file is `docs/supplemental-locked-test-before.txt` at `c5a999f5942b3220bd085c59b3f4d6e9a872d587`, timestamped October 14, 2026, 17:22:31 -04:00. Part B was blank in that commit; the BEFORE file remained unchanged before execution.

```
Human prediction date and time: 10/14 @ 5:18pm
Tested baseline Git commit: a2084eaddc466784f27a6187139615be08099d5f
Driver name and units: Consulting revenue-growth shift, percentage points
Old driver value: 0 percentage points
New driver value: -1 percentage points
Why the change is economically meaningful: Consulting is one of IBM's main operating segments and weaker demand
could reduce projected sales and profitability.
Expected revenue direction: Lower
Expected operating-profit direction: Lower
Expected FCFF direction: Lower
Expected intrinsic-value direction: Lower
Expected investment-decision effect: Watch/Defer
Reason for the prediction: Lower forecast operating cash flows would generally lower discounted operating value
for IBM.
```

Field	Verified result
Human prediction	At 10/14, 5:18pm, the student selected Consulting revenue-growth shift, 0pp to -1pp for FY2027-30, and predicted lower revenue, operating profit, FCFF, and intrinsic value, with Watch / Defer unchanged.
Execution	Started October 14, 2026, 17:25:03 -04:00. Existing API baseline, -1pp override, and 0pp reset were used.
Observed FY2030 change	Revenue $82,218.920m to $81,296.371m; EBIT $14,717.413m to $14,550.431m; FCFF $13,514.090m to $13,329.082m.
Value and decision	WACC unchanged at 7.070782%. Value $182.287630 to $179.793423/share, -$2.494207 (-1.368281%); below $231.40 and Watch / Defer unchanged.

Integrity and reset	Baseline, changed, and reset articulated PASS. Reset returned exactly $182.287630; low/high were $145.1346/$224.2811. The full suite passed 30/30.
Student evaluation	The student concluded the results matched because weaker Consulting growth reduced cash flows and discounted value, and retained Watch / Defer because $179.79 remained below $231.40 without sufficient margin of safety.

Verification: the pre-run Git commit, blank committed Part B, later execution timestamp, reset evidence, and objective results make this **VERIFIED** prospective precommit evidence. It does not retroactively verify the original October 10 prediction in the same way. See `docs/supplemental-locked-test-after.txt` for the full preserved Part A, Part B, and student-authored evaluation.

5. AI-use record and limitations

Exact model/version identifiers and individual interaction timestamps were not retained and are not claimed. Dates below are documented work dates or defensible October 2026 ranges. AI assistance is not itself independent evidence; the student owns the investment decision, approved assumptions, and final claims.

Tool	Material assistance	Disposition/check
ChatGPT — OpenAI	October 2026 planning, course interpretation, accounting/methodology discussions, source location, limitation review, independent-check planning, Codex instructions, written deliverables, Week 3 discovery, and compliance review.	Used as planning/drafting support; material claims checked against filings, model output, tests, and course sources.
Codex — OpenAI	October 2026 Python/debugging, workbench, tests, student-reviewed accounting corrections, documentation, HTML/PDF export, and Git operations under direction.	Validated through model output, 35 tests, PDF inspection, and student approval where required.
Published Week 3 benchmark	AI located, interpreted, and implemented the public Lab 05 benchmark.	Published inputs and expected outputs were consulted for required validation; calculations were generated from inputs and compared to the official reference. This was not access to private or unreleased grading answers.

The publicly available Week 3 Lab 05 training-case inputs and published expected outputs were consulted to complete the Project 1 known-answer reconciliation required by the course. AI assistance helped locate, interpret, and implement the benchmark. The calculated results were generated from the published inputs and compared against the official reference outputs. This was a disclosed validation exercise, not the use of private or unreleased grading answers.

- The September 1 timing correction from 122/184 to 121/184 was independently checked and enforced by tests.
- The student approved the October 14 SBC review; dedicated tests verify one CFO/APIC treatment and no operating-FCFF double count.
- The conventional Week 3 benchmark does not prove every IBM-specific timing, Financing, or ROIC-terminal convention.
- The $4.782bn gap is unresolved because company-defined FCF starts with consolidated CFO and adjusts Financing receivables/net capital investment, while operating FCFF is EBIT-based and excludes IBM Financing. Management outlook was not forced into the forecast; later cash-tax, Financing, working-capital, investment, or reconciliation disclosure can trigger re-underwriting.
- Favorable Software-growth cases remain below $231.40 under named causal scenarios. Qualified beta/debt-yield inputs, terminal-value dependence, future dilution, pension convention, and IBM Financing book-equity sidecar remain limitations.

6. Known-answer status

The published FIN 43900 Week 3 training case and answer table were located in the course repository. A standalone conventional-DCF implementation calculated all 12 outputs from published inputs; every comparison passed the documented tolerance. At 11% WACC, value was $23.4140/share, below the restored 10% baseline of $27.4974/share. See `docs/week3-known-answer-reconciliation.md` and `docs/known-answer-blocker.md` for sources, comparisons, commands, and scope.

This resolves the published Week 3 known-answer check. It does not make the training case an end-to-end test of IBM-specific timing, ROIC-terminal, Financing, or forecast logic. This PDF was generated from the corresponding HTML validation record. The Markdown source provides additional supporting detail.

7. Appendix A - Complete historical Locked Changed-Input AFTER record

The following AFTER record is reproduced verbatim from `docs/locked-changed-input-record-after.txt`. The separate BEFORE snapshot contains Part A's pre-run prediction with Part B blank. This experiment preceded the later SBC accounting repair; its $181.95 baseline and $190.31 observed value are not the current $182.29 base. Reported predecessor commit `7359ba2` is not independently verified here; archived BEFORE commit `4591725` is a later preservation event, not proof of the original prediction timestamp.

```
FIN 43900 PROJECT 1 – LOCKED CHANGED-INPUT RECORD
IBM Software Revenue Growth Test

PART A – BEFORE RUNNING THE TEST (AI OFF)

Date and time of prediction: 10/10/2026 @ 12:39pm
Baseline model Git commit ID: f91410f
Prediction-file Git commit ID (save after committing, before running): 4591725

Input being changed: Software revenue growth shift
Original input: 0 percentage points
New input: +2 percentage points (for 2027–2030)
Original IBM intrinsic value: $181.95 per share

MY PREDICTION (before running the test):
Expected effect on revenue, operating profit, and FCFF: I expect IBM's revenue, operating profit, and free cash
flow to increase as stronger software growth generates additional sales and earnings over the forecast period.

Expected effect on intrinsic value per share: I expect IBM's intrinsic value to increase above the baseline
valuation of $181.95 per share because higher projected free cash flows should increase the present value of the
business.

Expected effect on my investment recommendation: I expect IBM's valuation to become more attractive, but I do not
anticipate this adjustment alone will change my Watch/Defer recommendation, given the current market price of
$231.40.

Why I expect these changes: I believe IBM's AI and hybrid-cloud businesses could drive stronger software revenue
growth than my baseline forecast assumes. Higher software demand could improve operating earnings and free cash
flow, supporting a higher intrinsic valuation over time.


PART B – AFTER RUNNING THE TEST

Test date and time: 10/10/2026 @ 12:53pm
New IBM intrinsic value per share: $190.31
New 2030 operating profit (EBIT), $ millions: $15,279.2M
New 2030 FCFF, $ millions: $14,152.8M
WACC after change: 7.07%

Did the results move as I predicted?: Yes, as increasing IBM's software revenue growth by 2 percentage points
raised its intrinsic value from $181.95 to $190.31 per share,(+ $8.36 (4.6%)). Operating profit and free cash flow
also increased, which supports my prediction that stronger software growth would improve IBM's financial
performance and valuation.

Did my investment recommendation change?: No. My recommendation remains Watch/Defer. Although the higher software
growth assumption increased IBM's intrinsic value to $190.31, it remains below the market price of $231.40. This
suggests stronger growth alone is not enough to justify initiating a position at the current valuation.
```

If results differed from my prediction?: The results moved in the direction I expected, so there was no directional difference from my prediction. However, the valuation increase was not large enough to change my investment recommendation. This shows that even with stronger software growth, IBM's current market price remains above my model's estimated intrinsic value.

Location of before/after saved results: Screenshots of the original baseline ($181.95/share) and the +2 percentage-point software growth test ($190.31/share) are saved in my project’s GitHub repository.
Did I reset the model to its original baseline? Yes. I reset the software growth adjustment to 0 percentage points and restored the original baseline model.